Multilevel Models I

Math 243

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Outline

In today's class, we will...

- Introduce Multilevel Models (MLMs) as a method of accounting for "clustering" in data
- Discuss two kinds of MLMs: Random Intercepts and Random Slopes

Section 1

Multilevel Model Framework

Violating Regression Assumptions

Recall the standard linear regression model,

$$Y_i = \beta_0 + \beta_1 X_{1i} + \ldots, \beta_2 X_{2i} + \epsilon_i, \qquad \epsilon_i \sim N(0, \sigma^2)$$

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But we don't (yet) have a way to address violations of Independence!

- We'll propose a model to address one kind of non-independence: **Clustered Data**
- (There are others, e.g., time series data)

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- e.g., country-year "panel" data (World Bank Group, 2022)

Country	Year	Population
USA	2018	326.8 mil
USA	2019	328.3 mil
USA	2020	329.5 mil
Cameroon	2018	25.2 mil
Cameroon	2019	25.9 mil
Cameroon	2020	26.5 mil

"Clustered" data

School	Student	Grade	Hours Studied (per week)
School A	Mateo	92%	5
School A	Leslie	96%	6
School B	Bob	84%	8
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$\implies$ Students from **same** school likely have more similar grades than with students at **other** schools!

Potential Implications of Clustering

SLR Model: $\text{Grade} = \beta_0 + \beta_1 \text{Hours} + \epsilon$

1) Maybe we should have different intercepts for each school (or different *slopes* on Hours?)

$$\text{Grade} = \beta_{0,g} + \beta_1 \text{Hours} + \epsilon$$

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- We can do this by adding `School` as a predictor, i.e.,

$$\text{Grade} = \beta_0 + \beta_1 (\text{SchoolB}) + \beta_2 \text{Hours} + \epsilon$$

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2) Errors ϵ are not independent!

- Adding `School` as a predictor doesn't fix this!!

Multilevel Models

Multilevel Models (MLMs) are an extension of the linear model that gets at these two issues by:

1) Modeling different intercepts/slopes in each group (e.g., school)
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These models have many names:

- multilevel models
- hierarchical models
- mixed-effect models
- random effects models

Multilevel Models

Using previous example, the *most common* MLM assumes

$$\text{Grade} = \beta_0 + \beta_1 \text{Hours} + \underbrace{\gamma_g + \eta}_{\text{Error, } \epsilon}$$

where:

1. γ_g ("gamma") is a **random** school-varying term, often assumed that $\gamma_g \sim N(0, \sigma^2_\gamma)$
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Thus, we can write

$$\text{Grade} = \left(\beta_0 + \gamma_g\right) + \beta_1 \text{Hours} + \eta$$
$$\gamma_g \sim N(0, \sigma^2_\gamma)$$
$$\eta \sim N(0, \sigma^2_\eta)$$

1. Different Intercepts: The intercept specific to School g is $\beta_0 + \gamma_g$
2. Within-group Dependence: Errors $\epsilon = \gamma_g + \eta$ from same school depend on each other because they share γ_g

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- kth Predictor value: $x_{ki} \rightarrow x_{k,g[i]}$ (variable k for student i at school g)

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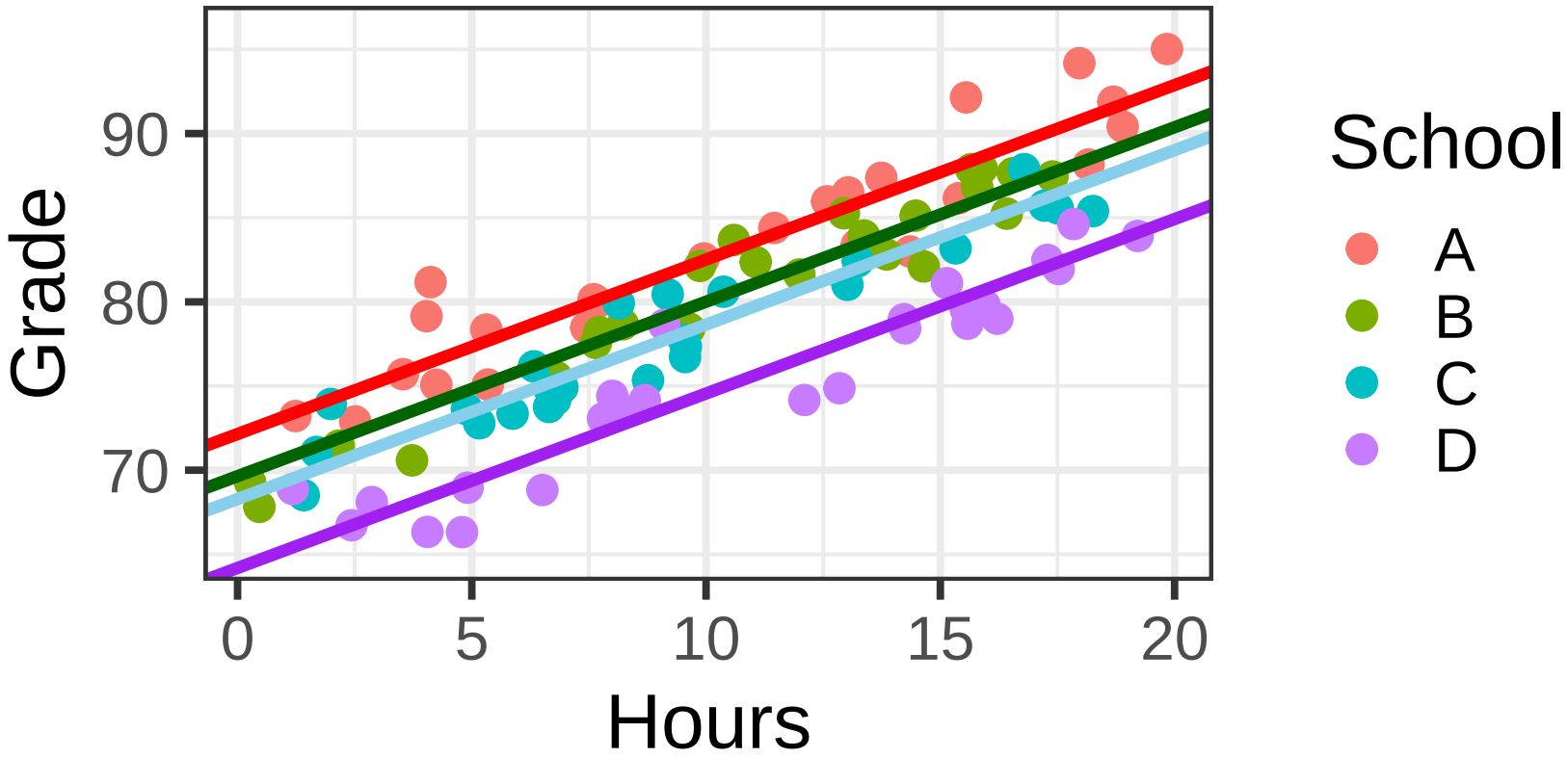
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- β_1: the slope on X
- σ^2_γ and σ^2_η: variance terms

Visualizing Random Intercept Model



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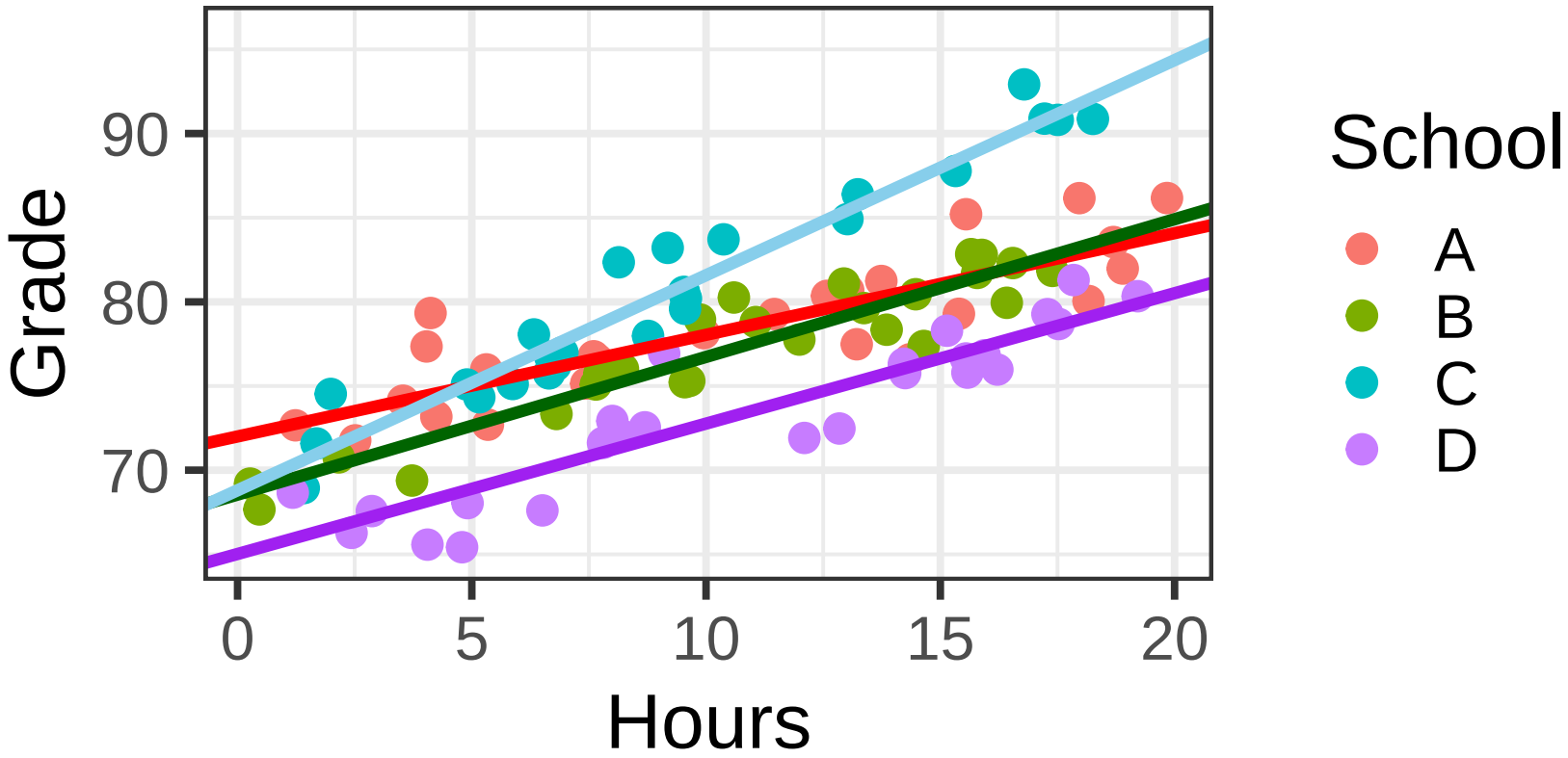
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Visualizing Random Slopes Model



Appears similar to previous "interaction" (fixed effects) models. . . but now we're **accounting for dependence!**

Benefits of Multilevel Models

1. **Explicitly account for dependence among correlated observations**
   - Vary intercepts by group ("random intercepts")
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3. **Under the hood, error estimates are more accurate $\Longrightarrow$ accurate inferences!**
   - If we don't account for correlation, confidence intervals, p-values would be wrong!
   - We can rely on inferences provided by `R`

Section 2

Think-Pair-Share Activity

Think-Pair-Share

Consider the Random Intercepts Model:

$$\begin{aligned} \text{Grade}_{g[i]} &= \beta_0 + \beta_1 \text{Hours}_{g[i]} + \gamma_g + \eta_{g[i]} \\ \gamma_g &\sim N(0, \sigma^2_\gamma) \\ \eta_{g[i]} &\sim N(0, \sigma^2_\eta) \end{aligned}$$

1. What does β_0 represent? β_1? Interpret each in a sentence, **in context**.
2. Suppose I told you $\beta_0 = 70$, $\beta_1 = 1$, and $\gamma_A = 10$. Make a prediction of the grade for a student at School A who studied $X = 15$ hours.
3. Suppose you wanted to make a prediction for a student at a *new* school that's **not in your dataset** (i.e., school Z), who studied $X = 15$ hours. What would you do?
4. Could you **ever** accurately model grades across schools using "fixed effects"? Compare and contrast "fixed effects" and "random intercepts".

Think-Pair-Share (Key)

1. **What does β_0 represent? β_1? Interpret each in a sentence, in context.**

β_0 is the expected grade for a student who studies 0 hours, on average. For every one additional hour of studying, we expect grade to increase by β_1, on average.

2. **Suppose I told you $\beta_0 = 70$, $\beta_1 = 1$, and $\gamma_A = 10$. Make a prediction of the grade for a student at School A who studied $X = 15$ hours.**

$$\beta_0 + \beta_1(15) + \gamma_A = 70 + 1(15) + 10 = 95$$

Think-Pair-Share (Key)

3. **Suppose you wanted to make a prediction for a student at a *new* school that's **not in your dataset** (i.e., school Z), who studied $X = 15$ hours. What would you do?**

I don't know γ_Z, so I'll just assume it's an "average" school, i.e., $\gamma_Z \approx 0$. Thus,

$$\beta_0 + \beta_1(15) + \gamma_Z = 70 + 1(15) + 0 = 85$$

4. **Could you **ever** accurately model grades across schools using "fixed effects"? Compare and contrast "fixed effects" and "random intercepts".**

Fixed effects could be accurate only if students within schools were independent. That's unrealistic, since students are social beings who know/work with each other!

Fixed effects and random effects both can model differences across group. But only random effects allows for correlation *within* groups.

Section 3

Recap

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Today, we...

- Addressed correlation (i.e., violations in independence) in linear regression with MLMs
- Discussed two kinds of MLMs: Random Intercepts and Random Slopes

Next time, we will...

- Implement Multilevel Models in R